

On the Local Optimality of Gaussian distributions for the Han-Kobayashi Inner Bound for the Gaussian Z-interference channel

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Abstract

We study the local optimality of Gaussian distributions for the Han-Kobayashi inner bound of the Gaussian Z-interference channel. We identify parameters (a, P_1, P_2) where Gaussian distributions do not optimize the Han-Kobayashi achievable region, as long as power control is not employed.

I. INTRODUCTION

In this paper, we study the scalar one-sided interference channel (or Z-interference) given by $Y_1 = X_1 + Z_1$ and $Y_2 = X_2 + X_1 + Z_2$, as depicted in Figure 1. Here, X_1 and X_2 are transmitted signals constrained to have average powers P_1 and P_2 , respectively, $a \in (0, 1)$ is an interference gain, Z_1 and Z_2 are standard Gaussians, and Y_1 and Y_2 are the two received signals. Thus, this Z-interference channel model is specified using three parameters (a, P_1, P_2) . We use the standard definition of the capacity region, denoted by $\mathcal{C}$, as considered in [1].

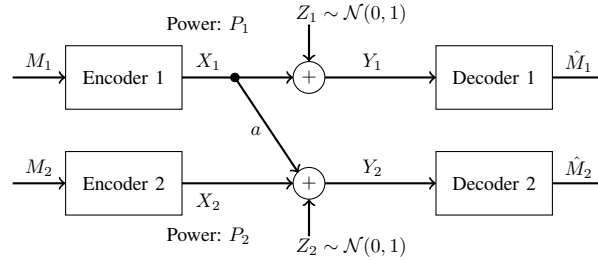


Fig. 1. Gaussian Z-interference Channel.

A. Han-Kobayashi achievable region

The most well-known inner bound for the interference channel is the Han-Kobayashi achievable region. while there are examples where the Han-Kobayashi inner bound is not equal to the capacity region [2], for certain special classes of interference channels, the Han-Kobayashi inner bound has been shown to be optimal [3], [4]. It remains an open question whether the Han-Kobayashi inner bound with Gaussian signaling and power control is tight for the Gaussian Z-interference channels.

Remark 1. The authors are aware of a recent claim, [5], along these lines. This work has primarily arisen from the authors' attempt to understand the various claims made in [5]. Please see Remark 2 in this regard.

Theorem 1 (HK). *Given a Gaussian Z-interference channel $p(y_1|x_1)p(y_2|x_1, x_2)$ with parameters (a, P_1, P_2) , a rate pair (R_1, R_2) is achievable if*

$$R_1 < I(X_1; Y_1|Q), \quad (1a)$$

$$R_2 < I(X_2; Y_2|U_1, Q), \quad (1b)$$

$$R_1 + R_2 < I(U_1, X_2; Y_2|Q) + I(X_1; Y_1|U_1, Q), \quad (1c)$$

for some $p(q)p(u_1, x_1|q)p(x_2|q)$ satisfying

$$E[|X_1|^2] \leq P_1, \quad (1d)$$

$$E[|X_2|^2] \leq P_2. \quad (1e)$$

We will call the closure of this region $\mathcal{R}_{HK}$.

The Han-Kobayashi inner bound, with Gaussian signaling, reduces to the following three inequalities for a Gaussian Z-interference channel.

Theorem 2 (HK-GS). *The union of rate pairs (R_1, R_2) satisfying*

$$\begin{aligned} R_1 &\leq \sum_{q=1}^{|\mathcal{Q}|} \frac{p(q)}{2} \log(1 + P_{1q}), \\ R_2 &\leq \sum_{q=1}^{|\mathcal{Q}|} \frac{p(q)}{2} \log \left(1 + \frac{P_{2q}}{1 + \alpha_q a^2 P_{1q}} \right), \\ R_1 + R_2 &\leq \sum_{q=1}^{|\mathcal{Q}|} \frac{p(q)}{2} \left(\log \left(1 + \frac{P_{2q} + (1 - \alpha_q) a^2 P_{1q}}{1 + \alpha_q a^2 P_{1q}} \right) + \log(1 + \alpha_q P_{1q}) \right), \end{aligned}$$

over $\alpha_q \in [0, 1]$, p_Q , and non-negative pairs P_{1q}, P_{2q} satisfying $\sum_q p(q) P_{1q} \leq P_1$ and $\sum_q p(q) P_{2q} \leq P_2$. By Caratheodory's theorem, we can restrict $|\mathcal{Q}| \leq 5$. We will call this region $\mathcal{R}_{HK-GS}$.

The primary conjecture that one wishes to resolve is the following:

Conjecture 1. *For the Gaussian Z-interference channel, Han-Kobayashi region with Gaussian Signaling coincides with the capacity region, i.e., $\mathcal{C} = \mathcal{R}_{HK-GS}$.*

A further restriction of the Han-Kobayashi region happens when we set $|\mathcal{Q}| = 1$. We call this the superposition coding region.

Theorem 3 (Sup). *Given a Gaussian Z-interference channel $p(y_1|x_1)p(y_2|x_1, x_2)$ with parameters (a, P_1, P_2) , a rate pair (R_1, R_2) is achievable if*

$$R_1 < I(X_1; Y_1), \quad (2a)$$

$$R_2 < I(X_2; Y_2|U_1), \quad (2b)$$

$$R_1 + R_2 < I(U_1, X_2; Y_2) + I(X_1; Y_1|U_1), \quad (2c)$$

for some $p(u_1, x_1)p(x_2)$ satisfying

$$E[|X_1|^2] \leq P_1, \quad (2d)$$

$$E[|X_2|^2] \leq P_2. \quad (2e)$$

We will call the closure of this region $\mathcal{R}_{Sup}$.

Similarly, one can also consider restricting the superposition coding region to Gaussian signaling to obtain the following achievable region.

Proposition 1 (Sup-GS). *The union of rate pairs (R_1, R_2) satisfying*

$$\begin{aligned} R_1 &\leq \frac{1}{2} \log(1 + P_1), \\ R_2 &\leq \frac{1}{2} \log \left(1 + \frac{P_2}{1 + \alpha a^2 P_1} \right), \\ R_1 + R_2 &\leq \frac{1}{2} \left(\log \left(1 + \frac{P_2 + (1 - \alpha) a^2 P_1}{1 + \alpha a^2 P_1} \right) + \log(1 + \alpha P_1) \right), \end{aligned} \quad (3)$$

over $\alpha \in [0, 1]$. We will call this region $\mathcal{R}_{Sup-GS}$.

Since the capacity region is a convex region in $\mathbb{R}_+^2$, it can be characterized by supporting hyperplanes of the form $R_1 + \beta R_2$. Therefore, we are interested in determining $\mathcal{C}^{(a, P_1, P_2)}(\beta) := \max_{(R_1, R_2) \in \mathcal{C}} R_1 + \beta R_2$. In this paper, we restrict to $\beta \geq 1$. For $0 \leq \beta \leq 1$, the value has been established in [6], [7].

Similarly, we can define $\mathcal{S}_{HK}^{(a, P_1, P_2)}(\beta)$, $\mathcal{S}_{HK-GS}^{(a, P_1, P_2)}(\beta)$, $\mathcal{S}_{Sup}^{(a, P_1, P_2)}(\beta)$ and $\mathcal{S}_{Sup-GS}^{(a, P_1, P_2)}(\beta)$ as the corresponding optimal weighted sum-rate for the corresponding regions. The following relationship between the quantities are immediate:

$$\mathcal{S}_{HK}^{(a, P_1, P_2)}(\beta) = \mathfrak{C}_{P_1, P_2}[\mathcal{S}_{Sup}^{(a, P_1, P_2)}(\beta)]; \quad \mathcal{S}_{HK-GS}^{(a, P_1, P_2)}(\beta) = \mathfrak{C}_{P_1, P_2}[\mathcal{S}_{Sup-GS}^{(a, P_1, P_2)}(\beta)], \quad (4)$$

where the concave envelopes are with respect to the power constraints, attained by the use of $\mathcal{Q}$.

B. Motivation

It is known [8] that there are parameters (a, P_1, P_2) where $\mathcal{R}_{HK-GS} \supsetneq \mathcal{R}_S$, or in other words, a non-trivial Q improves the achievable rate region with Gaussian signaling. However, all the converse techniques known to the authors, for proving the optimality of Gaussian signaling, do not allow for the presence of a non-trivial Q .

One potential way to get around this impasse is to perhaps show that $S_{Sup-GS}^{(a, P_1, P_2)}(\beta) = S_{Sup}^{(a, P_1, P_2)}(\beta)$, and show that it also holds for the multi-letter extensions of the channel. This would then establish Conjecture 1 due to the following reasoning. It is known, using Fano's inequality, that the $S_{Sup}^{(a, P_1, P_2)}(\beta)$ evaluated for the multi-letter extension goes to $C^{(a, P_1, P_2)}(\beta)$, and it has been established in [9] that the multi-letter extension of $S_{Sup-GS}^{(a, P_1, P_2)}(\beta)$ goes to $S_{HK-GS}^{(a, P_1, P_2)}(\beta)$.

Remark 2. A result submission in arXiv, [5], puts forward an approach similar to the one described above to claim a proof of Conjecture 1. However, the authors have been unable to verify some parts of the proof. The result in this paper establishes that there are parameters (a, P_1, P_2) , as well as β 's of interest where $S_{Sup-GS}^{(a, P_1, P_2)}(\beta) < S_{Sup}^{(a, P_1, P_2)}(\beta)$.

Let us define the following two critical slopes:

$$\begin{aligned}\beta_{Sato}(\mathcal{C}) &:= \sup \left\{ \beta : C^{(a, P_1, P_2)}(\beta) = \frac{1}{2} \log(1 + P_1) + \frac{\beta}{2} \log \left(1 + \frac{P_2}{1 + a^2 P_1} \right) \right\} \\ \beta_{CPW}(\mathcal{C}) &:= \inf \left\{ \beta : C^{(a, P_1, P_2)}(\beta) = \frac{1}{2} \log \left(1 + \frac{a^2 P_1}{1 + P_2} \right) + \frac{\beta}{2} \log(1 + P_2) \right\}.\end{aligned}$$

These correspond to the slopes of the capacity region at its two non-trivial corner points to the capacity region, called the Sato's corner point and the Costa-Polyanskiy-Wu corner point. Lower and Upper bounds on $\beta_{Sato}(\mathcal{C})$ and $\beta_{CPW}(\mathcal{C})$, are known in literature. (For instance, see [10] where the first non-trivial upper bound to $\beta_{CPW}(\mathcal{C})$ was established.)

In the same way, for each of the four achievable regions described above, we can define

$$\beta_{Sato}(\mathcal{R}_{HK}), \beta_{Sato}(\mathcal{R}_{HK-GS}), \beta_{Sato}(\mathcal{R}_{Sup}), \beta_{Sato}(\mathcal{R}_{Sup-GS}),$$

as well as

$$\beta_{CPW}(\mathcal{R}_{HK}), \beta_{CPW}(\mathcal{R}_{HK-GS}), \beta_{CPW}(\mathcal{R}_{Sup}), \beta_{CPW}(\mathcal{R}_{Sup-GS}).$$

The values of $\beta_{Sato}(\mathcal{R}_{HK-GS})$ and $\beta_{CPW}(\mathcal{R}_{HK-GS})$ have been established in [11], and [12] respectively. The values of $\beta_{Sato}(\mathcal{R}_{Sup-GS})$ and $\beta_{CPW}(\mathcal{R}_{Sup-GS})$ are simple exercises in elementary calculus (also established later in this paper).

II. MAIN

We formulate our main result in the following theorem.

Theorem 4. *Consider the Gaussian Z-interference channel specified by parameters (a, P_1, P_2) that satisfy $a^2(1 + P_1) > P_2 + \sqrt{P_2(P_2 + 1 - a^2)}$. There exists some $\beta \in (\beta_{Sato}(\mathcal{R}_{Sup-GS}), \beta_{CPW}(\mathcal{R}_{Sup-GS}))$ such that $S_{Sup-GS}^{(a, P_1, P_2)}(\beta) < S_{Sup}^{(a, P_1, P_2)}(\beta)$. In other words, Gaussians do not optimize the weighted sum-rate of the region $\mathcal{R}_{Sup}$ for some slopes between the two corner points of $\mathcal{R}_{Sup-GS}$.*

We use a perturbation-based approach to disprove the Gaussian optimality. As mentioned in literature, for instance in [13], perturbations using Hermite polynomials are very useful when exploring the neighbourhood of Gaussian distributions. In the next section, we recap some elementary properties of Hermite polynomials. In the spirit of being self-contained, we give a proof in Appendix A.

A. Hermite perturbation of Gaussian distributions

Define the following degree k -polynomial.

$$H_k^{(P)}(t) = \begin{cases} \frac{(-1)^k P^{\frac{k}{2}}}{\sqrt{k!}} e^{\frac{t^2}{2P}} \frac{d^k}{dt^k} e^{-\frac{t^2}{2P}}, & k \geq 1, \\ 1, & k = 0. \end{cases}$$

Let $g^{(P)}(t) = \frac{1}{\sqrt{2\pi P}} e^{-\frac{t^2}{2P}}$, denote the density of $\mathcal{N}(0, P)$. Define an inner product according to

$$\langle f_1, f_2 \rangle_P := \int_{-\infty}^{+\infty} f_1(t) f_2(t) g^{(P)}(t) dt.$$

The family of Hermite polynomials satisfies an orthogonality stated below (see Szego's standard book on Orthogonal Polynomials)

Lemma 1. *The family of polynomials $H_k^{(P)}(t)$ are orthonormal with respect to the inner product $\langle \cdot, \cdot \rangle_P$.*

Consider a perturbation of Gaussian distribution using Hermite polynomials:

$$g_\epsilon^{(P)}(t) = g^{(P)}(t) \left(1 + \epsilon \left(\sum_{k=1}^m a_k H_k^{(P)}(t) + \delta H_{2m}^{(P)}(t) \right) \right),$$

for some $\{a_k\}_{k=1}^m$ and $\delta > 0$, where $g^{(P)}(t)$ is the density of a Gaussian with mean zero and variance P .

Lemma 2. *The following hold:*

1) *The mean and second moment of such a perturbed distribution is given by*

$$\begin{aligned} \int_{-\infty}^{\infty} t g_\epsilon^{(P)}(t) dt &= \epsilon \sqrt{P} a_1 && \text{(by orthogonality)} \\ \int_{-\infty}^{\infty} t^2 g_\epsilon^{(P)}(t) dt &= P + \epsilon \sqrt{2P} a_2 && \text{(by orthogonality).} \end{aligned}$$

2) *The entropy of a perturbed Gaussian is given by*

$$- \int g_\epsilon^{(P)}(t) \log g_\epsilon^{(P)}(t) dt = - \int g^{(P)}(t) \log g^{(P)}(t) dt + \frac{\epsilon}{\sqrt{2}} a_2 - \frac{\epsilon^2}{2} \sum_{k \geq 1} a_k^2 + O(\epsilon^3).$$

Equivalently, there is a constant C and $\epsilon_0 > 0$ such that for all $\epsilon \in (0, \epsilon_0)$ we have

$$\left| - \int g_\epsilon^{(P)}(t) \log g_\epsilon^{(P)}(t) dt + \int g^{(P)}(t) \log g^{(P)}(t) dt - \frac{\epsilon}{\sqrt{2}} a_2 + \frac{\epsilon^2}{2} \sum_{k \geq 1} a_k^2 \right| < C \epsilon^3.$$

3) *If $X \sim g_\epsilon^{(P)}(t)$ then $aX \sim g_\epsilon^{(a^2 P)}(t)$.*

4) *If $X \sim g_\epsilon^{(P)}(t)$ and $Z \sim g^{(Q)}(t)$ are independent random variables, then*

$$X + Z \sim g^{(P+Q)}(t) \left(1 + \epsilon \sum_{k \geq 1} a_k \left(\frac{P}{P+Q} \right)^{\frac{k}{2}} H_k^{(P+Q)}(t) \right).$$

5) *More generally, if $X \sim g_\epsilon^{(P)}(t)$ and $Y \sim \hat{g}_\epsilon^{(Q)}(t)$ are independent random variables, then*

$$\begin{aligned} X + Y &\sim g^{(P+Q)}(t) \left(1 + \epsilon \sum_{k \geq 1} \left(a_k \left(\frac{P}{P+Q} \right)^{\frac{k}{2}} + \hat{a}_k \left(\frac{Q}{P+Q} \right)^{\frac{k}{2}} \right) H_k^{(P+Q)}(t) \right. \\ &\quad \left. + \epsilon^2 \sum_{k \geq 2} \sum_{l=1}^{k-1} a_l \hat{a}_{k-l} \frac{P^{\frac{l}{2}} Q^{\frac{k-l}{2}} \sqrt{k!}}{(P+Q)^{\frac{k}{2}} \sqrt{l!} \sqrt{(k-l)!}} H_k^{(P+Q)}(t) \right) \end{aligned}$$

6) *If $\tilde{g}_\epsilon^{(P)}(t) = g_\epsilon^{(P)}(t) + \epsilon^2 g^{(P)}(t) \sum_{k \geq 1} b_k H_k^{(P)}(t)$, then*

$$- \int \tilde{g}_\epsilon^{(P)}(t) \log \tilde{g}_\epsilon^{(P)}(t) dt = - \int g_\epsilon^{(P)}(t) \log g_\epsilon^{(P)}(t) dt + \frac{\epsilon^2}{\sqrt{2}} b_2 + O(\epsilon^3),$$

The proof can be found in Appendix A.

B. Preliminaries about $\mathcal{R}_{Sup-GS}$

For, $\beta \geq 1$, we have

$$S_{Sup-GS}^{(a, P_1, P_2)}(\beta) = \max_{\alpha \in [0, 1]} \frac{1}{2} \log(a^2 P_1 + P_2 + 1) + \frac{(\beta - 1)}{2} \log(a^2 \alpha P_1 + P_2 + 1) + \frac{1}{2} \log(\alpha P_1 + 1) - \frac{\beta}{2} \log(a^2 \alpha P_1 + 1).$$

Equating the derivative (of the inner expression) with respect to α yields

$$\frac{(\beta - 1) a^2 P_1}{P_2 + a^2 \alpha P_1 + 1} + \frac{P_1}{\alpha P_1 + 1} - \frac{\beta a^2 P_1}{a^2 \alpha P_1 + 1},$$

which can be expressed as

$$\frac{P_1 \left(P_2 + 1 - a^2 - \beta a^2 P_2 - a^2 \alpha P_1 ((\beta - 1) P_2 - (1 - a^2)) \right)}{(\alpha P_1 + 1)(a^2 \alpha P_1 + 1)(P_2 + a^2 \alpha P_1 + 1)}. \quad (5)$$

Therefore, for all $\beta \geq \beta_{CPW}(\mathcal{R}_{Sup-GS}) = \frac{P_1+1-a^2}{a^2P_2}$, we have that the optimal $\alpha = 0$ and

$$\mathcal{S}_{Sup-GS}^{(a,P_1,P_2)}(\beta) = \frac{1}{2} \log \left(1 + \frac{a^2P_1}{1+P_2} \right) + \frac{\beta}{2} \log(1+P_2).$$

Similarly, for all $\beta \leq \beta_{Sato}(\mathcal{R}_{Sup-GS}) = \frac{(P_2+1-a^2)(a^2P_1+1)}{a^2P_2(P_1+1)}$, we have that the optimal $\alpha = 1$ and

$$\mathcal{S}_{Sup-GS}^{(a,P_1,P_2)}(\beta) = \frac{1}{2} \log(1+P_1) + \frac{\beta}{2} \log \left(1 + \frac{P_2}{1+a^2P_1} \right).$$

Therefore any $\beta \in (\beta_{Sato}, \beta_{CPW})$ is in one-to-one correspondence with $\alpha \in (0, 1)$, by setting the expression in (5) to be zero. In other words, $\beta \in (\beta_{Sato}, \beta_{CPW})$ can be expressed as

$$\beta = \frac{(P_2 + 1 - a^2)(a^2\alpha P_1 + 1)}{a^2P_2(\alpha P_1 + 1)}, \quad (6)$$

for some $\alpha \in (0, 1)$.

C. Proof of Theorem 4

Let $U_1 \sim \mathcal{N}(0, (1-\alpha)P_1)$, $V_1 \sim \mathcal{N}(0, \alpha P_1)$, and $X_2 \sim \mathcal{N}(0, P_2)$ be independent Gaussians. Observe that region $\mathcal{R}_{Sup-GS}$ is obtained by setting $X_1 = U_1 + V_1$, where U_1, V_1 are as defined above, into the constraints of $\mathcal{R}_{Sup}$. We will establish Theorem 4 by showing that there is a mean and variance preserving perturbation of the distributions U_1, V_1 , and X_2 that increases the weighted sum rate in $\mathcal{R}_{Sup}$, or in other words,

$$I(U_1; Y_2) + I(X_1; Y_1|U_1) + \beta I(X_2; Y_2|U_1).$$

Consider the following perturbed random variables $(U_{1,\epsilon}, V_{1,\epsilon}, X_{2,\epsilon})$ defined by their densities

$$\begin{aligned} g_{U_{1,\epsilon}}^{(\bar{\alpha}P_1)}(t) &= g_{U_1}^{(\bar{\alpha}P_1)}(t) \left(1 + \epsilon \left(a_3 H_3^{(\bar{\alpha}P_1)}(t) + \delta_a H_6^{(\bar{\alpha}P_1)}(t) \right) \right), \\ g_{V_{1,\epsilon}}^{(\alpha P_1)}(t) &= g_{V_1}^{(\alpha P_1)}(t) \left(1 + \epsilon \left(b_3 H_3^{(\alpha P_1)}(t) + \delta_b H_6^{(\alpha P_1)}(t) \right) \right), \\ g_{X_{2,\epsilon}}^{(P_2)}(t) &= g_{X_2}^{(P_2)}(t) \left(1 + \epsilon \left(c_3 H_3^{(P_2)}(t) + \delta_c H_6^{(P_2)}(t) \right) \right), \end{aligned}$$

where $g_{U_1}^{(\bar{\alpha}P_1)}, g_{V_1}^{(\alpha P_1)}, g_{X_2}^{(P_2)}$ are the density functions of the associated Gaussians. We assume that $\delta_a, \delta_b, \delta_c > 0$ (this guarantees the validity of the perturbation by ensuring non-negativity of the perturbed density function in a non-zero open interval of ϵ containing $\epsilon = 0$). We further assume that these perturbed random variables are still independent of each other and set $X_1 = U_{1,\epsilon} + V_{1,\epsilon}$. Note that as the perturbation does not include the first and second order Hermite polynomials, the mean and the variance of the individual random variables are preserved (see Lemma 2.)

Define

$$f(\epsilon) := I(U_{1,\epsilon}; Y_2) + I(X_1; Y_1|U_{1,\epsilon}) + \beta I(X_2; Y_2|U_{1,\epsilon}).$$

Note that $f(0) = \mathcal{S}_{Sup-GS}^{(a,P_1,P_2)}(\beta)$. Using Lemma 2, we can obtain that (the details of the calculations are in the Appendix),

$$\begin{aligned} f(\epsilon) - f(0) &= -\frac{\epsilon^2}{2} \left(\left(a_3 \left(\frac{a^2\bar{\alpha}P_1}{1+a^2P_1+P_2} \right)^{\frac{3}{2}} + b_3 \left(\frac{a^2\alpha P_1}{1+a^2P_1+P_2} \right)^{\frac{3}{2}} + c_3 \left(\frac{P_2}{1+a^2P_1+P_2} \right)^{\frac{3}{2}} \right)^2 \right. \\ &\quad + \left(\delta_a \left(\frac{a^2\bar{\alpha}P_1}{1+a^2P_1+P_2} \right)^3 + \delta_b \left(\frac{a^2\alpha P_1}{1+a^2P_1+P_2} \right)^3 + \delta_c \left(\frac{P_2}{1+a^2P_1+P_2} \right)^3 \right)^2 \\ &\quad + (\beta - 1) \left(b_3 \left(\frac{a^2\alpha P_1}{1+a^2\alpha P_1+P_2} \right)^{\frac{3}{2}} + c_3 \left(\frac{P_2}{1+a^2\alpha P_1+P_2} \right)^{\frac{3}{2}} \right)^2 \\ &\quad + (\beta - 1) \left(\delta_b \left(\frac{a^2\alpha P_1}{1+a^2\alpha P_1+P_2} \right)^3 + \delta_c \left(\frac{P_2}{1+a^2\alpha P_1+P_2} \right)^3 \right)^2 \\ &\quad + \left(b_3^2 \left(\frac{\alpha P_1}{\alpha P_1 + 1} \right)^3 + \delta_b^2 \left(\frac{\alpha P_1}{\alpha P_1 + 1} \right)^6 \right) \\ &\quad \left. - \beta \left(b_3^2 \left(\frac{a^2\alpha P_1}{a^2\alpha P_1 + 1} \right)^3 + \delta_b^2 \left(\frac{a^2\alpha P_1}{a^2\alpha P_1 + 1} \right)^6 \right) \right). \end{aligned}$$

Now we set $a_3 = 0, c_3 = -b_3 \left(\frac{a^2 \alpha P_1}{P_2} \right)^{\frac{3}{2}}$. In this case, if

$$\beta \left(\frac{a^2 \alpha P_1}{a^2 \alpha P_1 + 1} \right)^3 > \left(\frac{\alpha P_1}{\alpha P_1 + 1} \right)^3, \quad (7)$$

then the δ 's can be chosen small enough to make the entire term positive. We now substitute for β from (6). Thus, the perturbed distributions will improve the weighted sum-rate when

$$\left(\frac{a^2 \alpha P_1}{a^2 \alpha P_1 + 1} \right)^3 > \frac{a^2 P_2 (\alpha P_1 + 1)}{(P_2 + 1 - a^2)(a^2 \alpha P_1 + 1)} \left(\frac{\alpha P_1}{\alpha P_1 + 1} \right)^3,$$

or equivalently

$$\frac{a^4}{(a^2 \alpha P_1 + 1)^2} > \frac{P_2}{(P_2 + 1 - a^2)} \frac{1}{(\alpha P_1 + 1)^2}.$$

This is equivalent to

$$\frac{a^2(1 + \alpha P_1)}{\alpha a^2 P_1 + 1} > \sqrt{\frac{P_2}{P_2 + 1 - a^2}},$$

for some $\alpha \in [0, 1]$. Since LHS is increasing in α , then as long as (a, P_1, P_2) satisfies the above at $\alpha = 1$, i.e.,

$$\begin{aligned} 1 - \frac{1 - a^2}{a^2 P_1 + 1} &> \sqrt{\frac{P_2}{P_2 + 1 - a^2}} = 1 - \frac{\sqrt{P_2 + 1 - a^2} - \sqrt{P_2}}{\sqrt{P_2 + 1 - a^2}} \\ \Leftrightarrow \frac{\sqrt{P_2 + 1 - a^2} - \sqrt{P_2}}{\sqrt{P_2 + 1 - a^2}} &> \frac{1 - a^2}{a^2 P_1 + 1} \\ \Leftrightarrow \frac{1 - a^2}{\sqrt{P_2 + 1 - a^2}(\sqrt{P_2 + 1 - a^2} + \sqrt{P_2})} &> \frac{1 - a^2}{a^2 P_1 + 1} \\ \Leftrightarrow a^2 P_1 + 1 &> P_2 + 1 - a^2 + \sqrt{P_2(P_2 + 1 - a^2)} \\ \Leftrightarrow a^2(1 + P_1) &> P_2 + \sqrt{P_2(P_2 + 1 - a^2)}, \end{aligned}$$

there is some $\beta \in (\beta_{Sato}(\mathcal{R}_{Sup-GS}), \beta_{CPW}(\mathcal{R}_{Sup-GS}))$ such that $S_{Sup-GS}^{(a, P_1, P_2)}(\beta) < S_{Sup}^{(a, P_1, P_2)}(\beta)$, completing the proof of Theorem 4. $\square$

Remark 3. A natural question is whether one can enlarge the region of parameters in Theorem 4 where $S_{Sup-GS}^{(a, P_1, P_2)}(\beta) < S_{Sup}^{(a, P_1, P_2)}(\beta)$, by considering a more general family of perturbations using higher order Hermite polynomials.

$$\begin{aligned} g_{U_1, \epsilon}^{(\bar{\alpha} P_1)}(t) &= g_{U_1}^{(\bar{\alpha} P_1)}(t) \left(1 + \epsilon \left(\sum_{k=3}^m a_k H_k^{(\bar{\alpha} P_1)}(t) + \delta_a H_{2m}^{(\bar{\alpha} P_1)}(t) \right) \right), \\ g_{V_1, \epsilon}^{(\alpha P_1)}(t) &= g_{V_1}^{(\alpha P_1)}(t) \left(1 + \epsilon \left(\sum_{k=3}^m b_k H_k^{(\alpha P_1)}(t) + \delta_b H_{2m}^{(\alpha P_1)}(t) \right) \right), \\ g_{X_2, \epsilon}^{(P_2)}(t) &= g_{X_2}^{(P_2)}(t) \left(1 + \epsilon \left(\sum_{k=3}^m c_k H_k^{(P_2)}(t) + \delta_c H_{2m}^{(P_2)}(t) \right) \right). \end{aligned}$$

Since we preserve the mean and variance of the perturbations, we do not use the first and second order Hermite polynomials in the perturbations. It is shown in the Appendix C that this family do not change the region of parameters in Theorem 4.

D. Relationship with Conjecture 1

One may wonder how the results in Theorem 4 affect Conjecture 1. If it turned out that

$$S_{HK-GS}^{(a, P_1, P_2)}(\beta) = S_{Sup-GS}^{(a, P_1, P_2)}(\beta) < S_{Sup}^{(a, P_1, P_2)}(\beta),$$

then, it would immediately disprove Conjecture 1.

We have a partial resolution along these lines. Last year, the authors proved the following result that identified the regime of parameters (a, P_1, P_2) where $S_{HK-GS}^{(a, P_1, P_2)}(\beta) = S_{Sup-GS}^{(a, P_1, P_2)}(\beta)$, $\forall \beta \in (\beta_{Sato}(\mathcal{R}_{HK-GS}), \beta_{CPW}(\mathcal{R}_{HK-GS}))$.

Theorem 5 (Theorem 3, [14]). *Consider a Gaussian Z-interference channel with parameters (P_1, P_2, a) . The noiseberg region consists only of a pure superposition coding strategy (i.e. no time-sharing is required for any β -sum-rate) whenever*

$$\frac{(P_2 + 1 - a^2)(a^2 P_1 + 1)}{a^2 P_2 (P_1 + 1)} \leq \beta^*, \quad (8)$$

where β^* is the unique positive solution of $\psi(\beta) = 0$, where ψ is defined as

$$\begin{aligned} \psi(\beta) := & \beta \left(\log \left(1 + \frac{P_2}{1 + a^2 P_1} \right) - \frac{(1 - a^2) P_2}{(1 + a^2 P_1)(1 + a^2 P_1 + P_2)} \right) \\ & + \log \left(1 - \frac{a^2 P_2 (1 + P_1)}{(1 + a^2 P_1)(1 + a^2 P_1 + P_2)} \beta \right). \end{aligned}$$

Further, in that paper, we also established the following lemma.

Lemma 3 (Lemma 3, [14]). *Let (P_1, P_2, a) satisfy (8), then $P_2 \geq a^2(1 + P_1)$.*

On the other hand, if (P_1, P_2, a) satisfies the condition in 4, then we must have $a^2(1 + P_1) > P_2$.

III. CONCLUSION

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APPENDIX A

ELEMENTARY RESULTS ABOUT HERMITE POLYNOMIALS AND PERTURBATIONS USING THEM

A. Preliminaries

Recall:

$$H_k^{(P)}(t) = \begin{cases} \frac{(-1)^k P^{\frac{k}{2}}}{\sqrt{k!}} e^{\frac{t^2}{2P}} \frac{d^k}{dt^k} e^{-\frac{t^2}{2P}}, & k \geq 1, \\ 1, & k = 0. \end{cases}$$

Note that:

$$\begin{aligned} \frac{d}{dt} \left(e^{\frac{t^2}{2P}} \frac{d^k}{dt^k} e^{-\frac{t^2}{2P}} \right) &= e^{\frac{t^2}{2P}} \left(\frac{t}{P} \frac{d^k}{dt^k} e^{-\frac{t^2}{2P}} + \frac{d^{k+1}}{dt^{k+1}} e^{-\frac{t^2}{2P}} \right) \\ &= e^{\frac{t^2}{2P}} \left(\frac{t}{P} \frac{d^k}{dt^k} e^{-\frac{t^2}{2P}} + \frac{d^k}{dt^k} \left(\frac{-t}{P} e^{-\frac{t^2}{2P}} \right) \right) \\ &= e^{\frac{t^2}{2P}} \left(\frac{t}{P} \frac{d^k}{dt^k} e^{-\frac{t^2}{2P}} + \frac{-t}{P} \frac{d^k}{dt^k} e^{-\frac{t^2}{2P}} - \frac{k}{P} \frac{d^{k-1}}{dt^{k-1}} e^{-\frac{t^2}{2P}} \right) \\ &= -\frac{k}{P} e^{\frac{t^2}{2P}} \frac{d^{k-1}}{dt^{k-1}} e^{-\frac{t^2}{2P}}. \end{aligned}$$

This implies that

$$\frac{(-1)^k \sqrt{k!}}{P^{\frac{k}{2}}} \frac{d}{dt} H_k^{(P)}(t) = -\frac{k}{P} \frac{(-1)^{k-1} \sqrt{(k-1)!}}{P^{\frac{k-1}{2}}} H_{k-1}^{(P)}(t),$$

or equivalently

$$\frac{d}{dt} H_k^{(P)}(t) = \sqrt{\frac{k}{P}} H_{k-1}^{(P)}(t), \quad k \geq 1. \quad (9)$$

Observe that

$$\begin{aligned} g^{(P)}(t) H_k^{(P)}(t) &= \frac{(-1)^k P^{\frac{k-1}{2}}}{\sqrt{2\pi} \sqrt{k!}} \frac{d^k}{dt^k} e^{-\frac{t^2}{2P}} \\ &= \frac{(-1)^k P^{\frac{k}{2}}}{\sqrt{k!}} \frac{d^k}{dt^k} g^{(P)}(t). \end{aligned} \quad (10)$$

Taking the derivative with respect to t on both sides yields

$$\begin{aligned} \frac{d}{dt} \left(g^{(P)}(t) H_k^{(P)}(t) \right) &= \frac{(-1)^k P^{\frac{k-1}{2}}}{\sqrt{2\pi} \sqrt{k!}} \frac{d^{k+1}}{dt^{k+1}} e^{-\frac{t^2}{2P}} \\ &= -\sqrt{\frac{k+1}{P}} g^{(P)}(t) H_{k+1}^{(P)}(t). \end{aligned}$$

Therefore

$$g^{(P)}(t) H_k^{(P)}(t) = -\sqrt{\frac{P}{k}} \frac{d}{dt} \left(g^{(P)}(t) H_{k-1}^{(P)}(t) \right), \quad k \geq 1. \quad (11)$$

Now we repeatedly use the fact that if $f(t)g(t) \rightarrow 0$ as $|t| \rightarrow \infty$, then

$$\int_{-\infty}^{\infty} f'(t)g(t)dt + \int_{-\infty}^{\infty} f(t)g'(t)dt = 0. \quad (12)$$

B. Proof of Lemma 1

If $\ell = k = 0$, then observe that

$$\langle H_0(t), H_0(t) \rangle_P = \int_{-\infty}^{+\infty} g^{(P)}(t) dt = 1.$$

If $\ell = 0$, then for $k \geq 1$ observe that

$$\begin{aligned} \langle H_k(t), H_0(t) \rangle_P &= \int_{-\infty}^{+\infty} g^{(P)}(t) H_k^{(P)}(t) dt \\ &= \int_{-\infty}^{+\infty} \frac{(-1)^k P^{\frac{k-1}{2}}}{\sqrt{2\pi} \sqrt{k!}} \frac{d^k}{dt^k} e^{-\frac{t^2}{2P}} dt \end{aligned}$$

$$= 0.$$

Now, observe that, for $k, \ell \geq 1$

$$\begin{aligned} \langle H_k(t), H_\ell(t) \rangle_P &= \int_{-\infty}^{+\infty} g^{(P)}(t) H_k^{(P)}(t) H_\ell^{(P)}(t) dt \\ &\stackrel{(a)}{=} \int_{-\infty}^{+\infty} -\sqrt{\frac{P}{k}} \frac{d}{dt} \left(g^{(P)}(t) H_{k-1}^{(P)}(t) \right) H_\ell^{(P)}(t) dt \\ &\stackrel{(b)}{=} \int_{-\infty}^{+\infty} \sqrt{\frac{P}{k}} g^{(P)}(t) H_{k-1}^{(P)}(t) \frac{d}{dt} H_\ell^{(P)}(t) dt \\ &\stackrel{(c)}{=} \int_{-\infty}^{+\infty} \sqrt{\frac{P}{k}} g^{(P)}(t) H_{k-1}^{(P)}(t) \sqrt{\frac{\ell}{P}} H_{\ell-1}^{(P)}(t) dt \\ &= \sqrt{\frac{\ell}{k}} \langle H_{k-1}(t), H_{\ell-1}(t) \rangle_P. \end{aligned}$$

We are done by induction, as it reduces to one of the base cases established earlier.

C. Proof of Lemma 2

1) We note that the second and third Hermite polynomials are

$$\begin{aligned} H_1^{(P)}(t) &= \frac{t}{\sqrt{P}}, \\ H_2^{(P)}(t) &= \frac{1}{\sqrt{2}} \left(\frac{t^2}{P} - 1 \right). \end{aligned}$$

By Lemma 1,

$$\begin{aligned} \int_{-\infty}^{\infty} t g_\epsilon^{(P)}(t) dt &= \int_{-\infty}^{\infty} \sqrt{P} H_1^{(P)}(t) g_\epsilon^{(P)}(t) dt = \epsilon \sqrt{P} a_1, \\ \int_{-\infty}^{\infty} t^2 g_\epsilon^{(P)}(t) dt &= \int_{-\infty}^{\infty} (\sqrt{2} P H_2^{(P)}(t) + P) g_\epsilon^{(P)}(t) dt = \epsilon \sqrt{2} P a_2 + P. \end{aligned}$$

2) The entropy of the perturbed Gaussian is

$$\begin{aligned} & - \int g_\epsilon^{(P)}(t) \log g_\epsilon^{(P)}(t) dt \\ &= - \int g^{(P)}(t) \left(1 + \epsilon \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right) \right) \log g^{(P)}(t) \left(1 + \epsilon \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right) \right) dt \\ &= - \int g^{(P)}(t) \left(\log g^{(P)}(t) + \log \left(1 + \epsilon \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right) \right) \right) dt \\ &\quad - \epsilon \int g^{(P)}(t) \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right) \left(\log g^{(P)}(t) + \log \left(1 + \epsilon \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right) \right) \right) dt \\ &= - \int g^{(P)}(t) \left(\log g^{(P)}(t) + \epsilon \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right) - \frac{\epsilon^2}{2} \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right)^2 + O(\epsilon^3) \right) dt \\ &\quad - \epsilon \int g^{(P)}(t) \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right) \left(\log g^{(P)}(t) + \epsilon \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right) - \frac{\epsilon^2}{2} \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right)^2 + O(\epsilon^3) \right) dt \\ &\stackrel{(a)}{=} - \int g^{(P)}(t) \log g^{(P)}(t) dt + \frac{\epsilon^2}{2} \left(\sum_{k=1}^{2m} a_k^2 \right) + O(\epsilon^3) \\ &\quad - \epsilon \int g^{(P)}(t) \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right) \left(-\frac{t^2}{2P} - \frac{1}{2} \log(2\pi P) \right) dt - \epsilon^2 \int g^{(P)}(t) \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right) dt + O(\epsilon^3) \\ &\stackrel{(b)}{=} - \int g^{(P)}(t) \log g^{(P)}(t) dt - \frac{\epsilon^2}{2} \left(\sum_{k=1}^{2m} a_k^2 \right) - \epsilon \int g^{(P)}(t) \left(\sum_{k=1}^{2m} a_k H_k^{(P)}(t) \right) \left(-\frac{1}{\sqrt{2}} H_2^{(P)}(t) \right) dt + O(\epsilon^3) \end{aligned}$$

$$\stackrel{(c)}{=} - \int g^{(P)}(t) \log g^{(P)} dt - \frac{\epsilon^2}{2} \left(\sum_{k=1}^{2m} a_k^2 \right) + \frac{\epsilon}{\sqrt{2}} a_2 + O(\epsilon^3).$$

Here (a), (b) and (c) uses the orthogonality established in Lemma 1.

3) Let $X \sim g_\epsilon^{(P)}$ with density $g_\epsilon^{(P)}(t)$, and fix $a \neq 0$. The density of aX is

$$\begin{aligned} \frac{1}{|a|} g_\epsilon^{(P)}\left(\frac{t}{a}\right) &= \frac{1}{|a|} g^{(P)}\left(\frac{t}{a}\right) \left(1 + \epsilon \sum_{k=1}^{2m} a_k H_k^{(P)}\left(\frac{t}{a}\right)\right) \\ &= g^{(a^2 P)}(t) \left(1 + \epsilon \sum_{k=1}^{2m} a_k H_k^{(a^2 P)}(t)\right) \\ &= g_\epsilon^{(a^2 P)}(t), \end{aligned}$$

as $g^{(P)}(t/a) \frac{1}{|a|} = g^{(a^2 P)}(t)$ and the Hermite polynomials satisfy

$$H_k^{(P)}\left(\frac{t}{a}\right) = H_k^{(a^2 P)}(t).$$

4) Let $X \sim g_\epsilon^{(P)}$ and $Z \sim g^{(Q)}$ independent. The density of $X + Z$ is the convolution

$$\begin{aligned} f_{X+Z}(t) &= \int g_\epsilon^{(P)}(x) g^{(Q)}(t-x) dx \\ &= \int g^{(P)}(x) g^{(Q)}(t-x) dx + \epsilon \sum_{k \geq 1} a_k \int g^{(P)}(x) H_k^{(P)}(x) g^{(Q)}(t-x) dx \\ &\stackrel{(a)}{=} g^{(P+Q)}(t) + \epsilon \sum_{k \geq 1} a_k \int \frac{(-1)^k P^{\frac{k}{2}}}{\sqrt{k!}} \frac{d^k}{dx^k} g^{(P)}(x) g^{(Q)}(t-x) dx, \\ &\stackrel{(b)}{=} g^{(P+Q)}(t) + \epsilon \sum_{k \geq 1} a_k \frac{(-1)^k P^{\frac{k}{2}}}{\sqrt{k!}} \frac{d^k}{dt^k} g^{(P+Q)}(t) \\ &\stackrel{(c)}{=} g^{(P+Q)}(t) + \epsilon \sum_{k \geq 1} a_k \left(\frac{P}{P+Q} \right)^{\frac{k}{2}} g^{(P+Q)}(t) H_k^{(P+Q)}(t). \end{aligned} \tag{13}$$

Here (a), (c) follow from (10), while (b) follows from the commutative property of convolution and differentiation.

5) Since X, Y are independent, then the distribution of $X + Y$ is given by the convolution

$$X + Y \sim \int g_\epsilon^{(P)}(x) \hat{g}_\epsilon^{(Q)}(t-x) dx.$$

Expand both perturbations:

$$g_\epsilon^{(P)}(x) = g^{(P)}(x) \left(1 + \epsilon \sum_{l \geq 1} a_l H_l^{(P)}(x)\right), \quad \hat{g}_\epsilon^{(Q)}(t-x) = g^{(Q)}(t-x) \left(1 + \epsilon \sum_{m \geq 1} \hat{a}_m H_m^{(Q)}(t-x)\right).$$

Then

$$\begin{aligned} &g_\epsilon^{(P)} * \hat{g}_\epsilon^{(Q)}(t) \\ &= \int g^{(P)}(x) g^{(Q)}(t-x) dx + \epsilon \sum_{\ell \geq 1} a_\ell \int g^{(P)}(x) H_\ell^{(P)}(x) g^{(Q)}(t-x) dx \\ &\quad + \epsilon \sum_{m \geq 1} \hat{a}_m \int g^{(P)}(x) g^{(Q)}(t-x) H_m^{(Q)}(t-x) dx + \epsilon^2 \sum_{l, m \geq 1} a_l \hat{a}_m \int g^{(P)}(x) H_l^{(P)}(x) g^{(Q)}(t-x) H_m^{(Q)}(t-x) dx \\ &\stackrel{(a)}{=} g^{(P+Q)}(t) + \epsilon \sum_{\ell \geq 1} a_\ell \left(\frac{P}{P+Q} \right)^{\frac{\ell}{2}} g^{(P+Q)}(t) H_\ell^{(P+Q)}(t) + \epsilon \sum_{m \geq 1} \hat{a}_m \left(\frac{Q}{P+Q} \right)^{\frac{m}{2}} g^{(P+Q)}(t) H_m^{(P+Q)}(t) \\ &\quad + \epsilon^2 \sum_{\ell, m \geq 1} a_\ell \hat{a}_m \int \frac{(-1)^\ell P^{\frac{\ell}{2}}}{\sqrt{\ell!}} \frac{d^\ell}{dx^\ell} g^{(P)}(x) \frac{(-1)^m Q^{\frac{m}{2}}}{\sqrt{m!}} \frac{d^m}{dx^m} g^{(Q)}(t-x) \Big|_{t-x} dx \\ &\stackrel{(b)}{=} g^{(P+Q)}(t) + \epsilon \sum_{\ell \geq 1} a_\ell \left(\frac{P}{P+Q} \right)^{\frac{\ell}{2}} g^{(P+Q)}(t) H_\ell^{(P+Q)}(t) + \epsilon \sum_{m \geq 1} \hat{a}_m \left(\frac{Q}{P+Q} \right)^{\frac{m}{2}} g^{(P+Q)}(t) H_m^{(P+Q)}(t) \end{aligned}$$

$$\begin{aligned}
& + \epsilon^2 \sum_{\ell, m \geq 1} a_\ell \hat{a}_m \frac{(-1)^{\ell+m} P^{\frac{\ell+m}{2}}}{\sqrt{\ell!} \sqrt{m!}} \frac{d^{\ell+m}}{dt^{\ell+m}} g^{(P+Q)}(t) \\
& \stackrel{(c)}{=} g^{(P+Q)}(t) + \epsilon \sum_{\ell \geq 1} a_\ell \left(\frac{P}{P+Q} \right)^{\frac{\ell}{2}} g^{(P+Q)}(t) H_k^{(P+Q)}(t) + \epsilon \sum_{m \geq 1} \hat{a}_m \left(\frac{Q}{P+Q} \right)^{\frac{m}{2}} g^{(P+Q)}(t) H_k^{(P+Q)}(t) \\
& + \epsilon^2 \sum_{\ell, m \geq 1} a_\ell \hat{a}_m \frac{\sqrt{(\ell+m)!}}{\sqrt{\ell!} \sqrt{m!}} g^{(P+Q)}(t) H_{\ell+m}^{(P+Q)}(t).
\end{aligned}$$

In the above (a) and (c) follow from (10), while (b) follows from the commutative property of convolution and differentiation.

6) Denote

$$A(t) := \sum_{k=1}^{2m} a_k H_k^{(P)}(t), \quad B(t) := \sum_{k=1}^{2m} b_k H_k^{(P)}(t).$$

Then

$$\tilde{g}_\epsilon^{(P)}(t) = g^{(P)}(t)(1 + \epsilon A(t) + \epsilon^2 B(t)).$$

Recall that $g_\epsilon^{(P)}(t) = g^{(P)}(t)(1 + \epsilon A(t))$. Therefore,

$$\begin{aligned}
- \int \tilde{g}_\epsilon^{(P)}(t) \log \tilde{g}_\epsilon^{(P)}(t) dt &= - \int g^{(P)}(t) \cdot (1 + \epsilon A(t) + \epsilon^2 B(t)) \log(g^{(P)}(t) \cdot (1 + \epsilon A(t) + \epsilon^2 B(t))) dt \\
&= - \int g^{(P)}(t) \cdot (1 + \epsilon A(t) + \epsilon^2 B(t)) \log(g^{(P)}(t)(1 + \epsilon A(t))) dt \\
&\quad - \int g^{(P)}(t) \cdot (1 + \epsilon A(t) + \epsilon^2 B(t)) \log \left(1 + \frac{\epsilon^2 B(t)}{1 + \epsilon A(t)} \right) dt \\
&= - \int g_\epsilon^{(P)}(t) \log g_\epsilon^{(P)}(t) dt - \epsilon^2 \int g^{(P)}(t) B(t) \log g^{(P)}(t) dt \\
&\quad - \epsilon^2 \int g^{(P)}(t) B(t) dt + O(\epsilon^3).
\end{aligned}$$

As $\log g^{(P)}(t) = \left(-\frac{t^2}{2P} - \frac{1}{2} \log(2\pi P) \right) = -\frac{1}{\sqrt{2}} H_2^{(P)}(t) - \frac{1}{2} \log(2\pi e P) H_0^{(P)}(t)$, using Lemma 1, it is immediate that

$$- \int \tilde{g}_\epsilon^{(P)}(t) \log \tilde{g}_\epsilon^{(P)}(t) dt = - \int g_\epsilon^{(P)}(t) \log g_\epsilon^{(P)}(t) dt + \frac{\epsilon^2}{\sqrt{2}} b_2 + O(\epsilon^3).$$

APPENDIX B

CALCULATIONS OF THE PERTURBED FUNCTION

In this section, we provide the calculation for the function $f(\epsilon)$.

According to Theorem 3, since

$$\begin{aligned}
& I(U_1, X_2; Y_2) + I(X_1; Y_1 | U_1) + (\beta - 1) I(X_2; Y_2 | U_1) \\
& = h(X_2 + a(U_1 + V_1) + Z_2) + h(X_2 + aV_1 + Z_2) + h(V_1 + Z_2) - h(Z_1) - \beta h(aV_1 + Z_2),
\end{aligned}$$

we need to derive the entropies $h(X_{2,\epsilon} + a(U_{1,\epsilon} + V_{1,\epsilon}) + Z_2)$, $h(X_{2,\epsilon} + aV_{1,\epsilon} + Z_2)$, $h(V_{1,\epsilon} + Z_2)$ and $h(aV_{1,\epsilon} + Z_2)$.

First, for the distributions of $aU_{1,\epsilon}$, $aV_{1,\epsilon}$, we use part 3) of Lemma 2 to derive the following:

$$\begin{aligned}
g_{aU_{1,\epsilon}}^{(a^2 \bar{\alpha} P_1)}(t) &= g_{aU_1}^{(a^2 \bar{\alpha} P_1)}(t) \left(1 + \epsilon \left(a_3 H_3^{(a^2 \bar{\alpha} P_1)}(t) + \delta_a H_6^{(a^2 \bar{\alpha} P_1)}(t) \right) \right) \\
g_{aV_{1,\epsilon}}^{(a^2 \alpha P_1)}(t) &= g_{aV_1}^{(a^2 \alpha P_1)}(t) \left(1 + \epsilon \left(b_3 H_3^{(a^2 \alpha P_1)}(t) + \delta_b H_6^{(a^2 \alpha P_1)}(t) \right) \right),
\end{aligned}$$

Then we use part 4) of Lemma 2 to derive:

$$aV_1 + Z_2 \sim g^{(a^2 \alpha P_1 + 1)}(t) \left(1 + \epsilon \left(b_3 \left(\frac{a^2 \alpha P_1}{a^2 \alpha P_1 + 1} \right)^{\frac{3}{2}} H_3^{(a^2 \alpha P_1)}(t) + \delta_b \left(\frac{a^2 \alpha P_1}{a^2 \alpha P_1 + 1} \right)^3 H_6^{(a^2 \alpha P_1)}(t) \right) \right)$$

$$V_1 + Z_2 \sim g^{(\alpha P_1 + 1)}(t) \left(1 + \epsilon \left(b_3 \left(\frac{\alpha P_1}{\alpha P_1 + 1} \right)^{\frac{3}{2}} H_3^{(\alpha P_1)}(t) + \delta_b \left(\frac{\alpha P_1}{\alpha P_1 + 1} \right)^3 H_6^{(\alpha P_1)}(t) \right) \right)$$

By part 5),

$$X_2 + aV_1 + Z_2 \sim g^{(a^2\alpha P_1 + P_2 + 1)}(t) \left(1 + \epsilon \left(b_3 \left(\frac{a^2\alpha P_1}{1 + a^2\alpha P_1 + P_2} \right)^{\frac{3}{2}} + c_3 \left(\frac{P_2}{1 + a^2\alpha P_1 + P_2} \right)^{\frac{3}{2}} \right) H_3^{(P_2 + a^2\alpha P_1 + 1)}(t) \right. \\ \left. + \epsilon \left(\delta_b \left(\frac{a^2\alpha P_1}{1 + a^2\alpha P_1 + P_2} \right)^3 + \delta_c \left(\frac{P_2}{1 + a^2\alpha P_1 + P_2} \right)^3 \right) H_6^{(P_2 + a^2\alpha P_1 + 1)}(t) \right) + O(\epsilon^2),$$

where the ϵ^2 terms are in the span of $\{H_k^{(P_2 + a^2\alpha P_1 + 1)}(t)\}_{k \geq 3}$, and hence do not affect the leading perturbation term for entropy.

Finally, we have $X_2 + a(U_1 + V_1) + Z_2 \sim$

$$g^{(a^2 P_1 + P_2 + 1)}(t) \left(1 + \epsilon \left(a_3 \left(\frac{a^2\bar{\alpha} P_1}{1 + a^2 P_1 + P_2} \right)^{\frac{3}{2}} + b_3 \left(\frac{a^2\alpha P_1}{1 + a^2 P_1 + P_2} \right)^{\frac{3}{2}} + c_3 \left(\frac{P_2}{1 + a^2 P_1 + P_2} \right)^{\frac{3}{2}} \right) H_3^{(P_2 + a^2 P_1 + 1)}(t) \right. \\ \left. + \epsilon \left(\delta_a \left(\frac{a^2\bar{\alpha} P_1}{1 + a^2 P_1 + P_2} \right)^3 + \delta_b \left(\frac{a^2\alpha P_1}{1 + a^2 P_1 + P_2} \right)^3 + \delta_c \left(\frac{P_2}{1 + a^2 P_1 + P_2} \right)^3 \right) H_6^{(P_2 + a^2 P_1 + 1)}(t) \right) + O(\epsilon^2).$$

Using parts 2) and 6) of Lemma 2, we obtain the expression of $f(\epsilon) - f(0)$ as desired.

APPENDIX C

A GENERAL PERTURBATION FAMILY

Let us consider a family of perturbations on (U_1, V_1, X_2) parametrized by $\{a_k\}_{k=3}^{2m_a}$, $\{b_k\}_{k=3}^{2m_b}$, $\{c_k\}_{k=3}^{2m_c}$ and $\epsilon > 0$ with $2m_a, 2m_b, 2m_c > 0$ as follows:

$$g_{U_1, \epsilon}^{(\bar{\alpha} P_1)}(t) = g_{U_1}^{(\bar{\alpha} P_1)}(t) \left(1 + \epsilon \left(\sum_{k=3}^{2m_a} a_k H_k^{(\bar{\alpha} P_1)}(t) \right) \right), \\ g_{V_1, \epsilon}^{(\alpha P_1)}(t) = g_{V_1}^{(\alpha P_1)}(t) \left(1 + \epsilon \left(\sum_{k=3}^{2m_b} b_k H_k^{(\alpha P_1)}(t) \right) \right), \\ g_{X_2, \epsilon}^{(P_2)}(t) = g_{X_2}^{(P_2)}(t) \left(1 + \epsilon \left(\sum_{k=3}^{2m_c} c_k H_k^{(P_2)}(t) \right) \right).$$

Note that by Lemma 2 part 1), this family does not change the first and second moments of the original Gaussians.

Let $m = \max\{m_a, m_b, m_c\}$. Then follow the previous derivations,

$$g_{aU_1, \epsilon}^{(a^2\bar{\alpha} P_1)}(t) = g_{aU_1}^{(a^2\bar{\alpha} P_1)}(t) \left(1 + \epsilon \left(\sum_{k=3}^{2m} a_k H_k^{(a^2\bar{\alpha} P_1)}(t) \right) \right), \\ g_{aV_1, \epsilon}^{(a^2\alpha P_1)}(t) = g_{aV_1}^{(a^2\alpha P_1)}(t) \left(1 + \epsilon \left(\sum_{k=3}^{2m} b_k H_k^{(a^2\alpha P_1)}(t) \right) \right).$$

The distributions are given by

$$V_1 + Z_2 \sim g_{U_1}^{(\alpha P_1)}(t) \left(1 + \epsilon \left(\sum_{k=3}^{2m} b_k \left(\frac{\alpha P_1}{\alpha P_1 + 1} \right)^{\frac{k}{2}} H_k^{(\alpha P_1 + 1)}(t) \right) \right), \\ aV_1 + Z_2 \sim g_{aU_1}^{(a^2\alpha P_1)}(t) \left(1 + \epsilon \left(\sum_{k=3}^{2m} b_k \left(\frac{a^2\alpha P_1}{a^2\alpha P_1 + 1} \right)^{\frac{k}{2}} H_k^{(\alpha P_1 + 1)}(t) \right) \right) \\ X_2 + aV_1 + Z_2 \sim g^{(a^2\alpha P_1 + P_2 + 1)}(t) \left(1 + \epsilon \left(\sum_{k=3}^{2m} b_k \left(\frac{a^2\alpha P_1}{1 + a^2\alpha P_1 + P_2} \right)^{\frac{k}{2}} + c_k \left(\frac{P_2}{1 + a^2\alpha P_1 + P_2} \right)^{\frac{k}{2}} \right) H_k^{(a^2\alpha P_1 + P_2 + 1)}(t) \right) \\ X_2 + a(U_1 + V_1) + Z_2 \sim g^{(a^2 P_1 + P_2 + 1)}(t) \\ \left(1 + \epsilon \sum_{k=3}^{2m} \left(a_k \left(\frac{a^2\bar{\alpha} P_1}{1 + a^2 P_1 + P_2} \right)^{\frac{k}{2}} + b_k \left(\frac{a^2\alpha P_1}{1 + a^2 P_1 + P_2} \right)^{\frac{k}{2}} + c_k \left(\frac{P_2}{1 + a^2 P_1 + P_2} \right)^{\frac{k}{2}} \right) H_k^{(a^2 P_1 + P_2 + 1)}(t) \right).$$

Therefore,

$$\begin{aligned}
f(\epsilon) - f(0) = & -\frac{\epsilon^2}{2} \sum_{k=3}^{2m} \left(\left[a_k \left(\frac{a^2 \bar{\alpha} P_1}{1 + a^2 P_1 + P_2} \right)^{\frac{k}{2}} + b_k \left(\frac{a^2 \alpha P_1}{1 + a^2 P_1 + P_2} \right)^{\frac{k}{2}} + c_k \left(\frac{P_2}{1 + a^2 P_1 + P_2} \right)^{\frac{k}{2}} \right]^2 \right. \\
& + (\beta - 1) \left[b_k \left(\frac{a^2 \alpha P_1}{1 + a^2 \alpha P_1 + P_2} \right)^{\frac{k}{2}} + c_k \left(\frac{P_2}{1 + a^2 \alpha P_1 + P_2} \right)^{\frac{k}{2}} \right]^2 \\
& \left. + b_k^2 \left(\frac{\alpha P_1}{\alpha P_1 + 1} \right)^k - \beta b_k^2 \left(\frac{a^2 \alpha P_1}{a^2 \alpha P_1 + 1} \right)^k \right).
\end{aligned}$$

For $f(\epsilon) - f(0)$ to be positive, observe that it is necessary that

$$\sum_{k=3}^{2m} b_k^2 \left(\beta \left(\frac{a^2 \alpha P_1}{a^2 \alpha P_1 + 1} \right)^k - \left(\frac{\alpha P_1}{\alpha P_1 + 1} \right)^k \right) > 0.$$

This necessitates that

$$\beta \left(\frac{a^2 \alpha P_1}{a^2 \alpha P_1 + 1} \right)^k > \left(\frac{\alpha P_1}{\alpha P_1 + 1} \right)^k \tag{14}$$

must hold for some $k \geq 3$.

On the other hand, if

$$\beta \left(\frac{a^2 \alpha P_1}{a^2 \alpha P_1 + 1} \right)^3 > \left(\frac{\alpha P_1}{\alpha P_1 + 1} \right)^3,$$

then it is immediate that (14) holds for all $k \geq 3$. As can be observed from (7), this is exactly the same condition as induced by the limited perturbation family considered in Theorem 4.